

Path to Fold-Transversal Magic State Cultivation on Neutral Atom Hardware*

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Fold-Transversal Magic State Cultivation (FTMSC) is a recently proposed method for generating high-fidelity magic states for non-Clifford logical operations. Due to its reliance on long-range CCZ gates, this procedure is well suited for neutral atom hardware because neutral atom qubits can either be moved into interaction zones or interact via the Rydberg blockade over long distances. This work studies possible implementations of FTMSC on realistic neutral atom hardware through direct simulation of adiabatically found asymmetric CCZ gates, which are then passed into a `stim` simulation of the later steps of FTMSC. We compare the capabilities of in-place and atom-shuttling architectures while also considering how common error channels affect the logical error rates of the final magic states.

I. INTRODUCTION

Recently, a new method of magic state cultivation (MSC) was introduced in [1] referred to as “Fold-Transversal Magic State Cultivation.” Their proposed algorithm allows for up to a tenfold decrease in the expected number of attempts required to create a high-fidelity magic state over previous implementations such as [2]. One of the key steps in this protocol is the use of control-control-Z (CCZ) gates to make a non-Clifford logical check on the magic $|H_{XY}\rangle$ states. That check enables the post-selection and further growth steps. This proposed technique is recommended for qubit architectures that can properly use long-range interactions to entangle qubits [1]. Neutral atom qubits are a strong choice for use in Fold-Transversal Magic State Cultivation due to their natural ability to communicate over distance through atom movement or multi-site laser addressing [3, 4].

There have been previous attempts to create high-fidelity CCZ gates with success, such as in [5]. These gates were created under the assumption of infinite blockade strength and a triangular arrangement of atoms. Both of these design choices allowed for simpler gradient ascent pulse engineering (GRAPE) optimization of the pulses. CCZ gates with asymmetric geometries have not been explored as much in the literature. In order for FTMSC to be implemented on neutral atoms, many asymmetric CCZ gates will need to be found. Using GRAPE or other such expensive optimization techniques for each asymmetric coupling is possible, but finding an optimal pulse shape for each possible coupling through a full numerical optimization is costly [6]. Instead, we develop a new adiabatic algorithm to quickly find low-infidelity, time-optimal CCZ pulses for any given three-atom arrangement.

We examine the fidelity across different couplings as well as under common noise sources in neutral atoms such as Rydberg decay, atom motion, and laser intensity fluctuations. As these gates will be non-unitary due to

leakage, we adapt a quantum trajectories method into the handoff from statevector simulation to `stim` simulation found in the original FTMSC paper’s github implementation [7].

A. Neutral Atom Hamiltonian

We describe the neutral atoms as a three-level system made up of the states $|0\rangle$, $|1\rangle$, and $|r\rangle$. The states $|0\rangle$ and $|1\rangle$ are hyperfine electronic states representing the computational basis, and $|r\rangle$, the Rydberg state, is used to mediate the interactions between atoms [8, 9]. A laser tuned to the transition between $|1\rangle$ and $|r\rangle$ drives the dynamics through a Rabi oscillation between the $|1\rangle$ and $|r\rangle$ states and a detuning caused by the laser phase, ϕ , where $\delta_{1r} = \frac{d\phi}{dt}$. Working in the rotating-wave approximation and taking $\hbar = 1$, the Hamiltonian for a single atom is

$$H_i(t) = \frac{\Omega_{1r}(t)}{2} (|1_i\rangle \langle r_i| + |r_i\rangle \langle 1_i|) + \delta_{1r}(t) |r_i\rangle \langle r_i|. \quad (1)$$

Note that in this work we recognize that the maximum Rabi drive is dependent on the laser, and as such we normalize energy in units of $\Omega_{\max}$. Neutral atoms couple to each other through an inter-atomic interaction term referred to as the Rydberg blockade [8, 9],

$$H_B = \sum_{i < j} B_{ij} |r_i r_j\rangle \langle r_i r_j|, \quad (2)$$

which represents an energy shift by the blockade strength, B_{ij} , when atoms i and j are both excited to the Rydberg state.

The total Hamiltonian representing n qubits is then

$$H(t) = \sum_{i=1}^n H_i(t) + \sum_{i < j} B_{ij} |r_i r_j\rangle \langle r_i r_j|. \quad (3)$$

Our analysis in this work only concerns the case up to $n = 3$. The Rabi drive, $\Omega_{1r}(t)$, is implemented as a flat-top pulse with a small ramp-up time, τ . The detuning of this pulse is implemented as a Fourier-inspired ansatz,

$$\delta_{1r}(t) = \Delta - \sum_{i=1}^4 a_i \sin(\omega_i t' - \phi_i), \quad (4)$$

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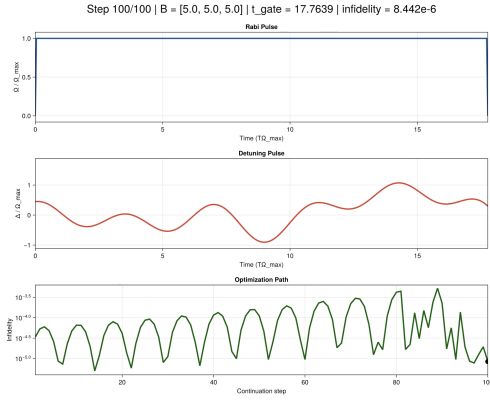


FIG. 1. Details of the time-optimal pulse used to implement a CCZ gate as well as the optimization path to get to a target blockade strength of $[5, 5, 5] V/\Omega_{\max}$.

where Δ is a constant detuning term and $t' = \frac{t-t_0}{2}$ with t_0 being the total pulse time. This form is chosen to give a flexible but still low-dimensional control parameterization for constructing the asymmetric CCZ gates considered in this work [5].

B. Fold-Transversal MSC

Fold-Transversal MSC is a recently published protocol for efficient preparation of magic states for use in non-Clifford logical operations. The protocol prepares a high-fidelity logical eigenstate of a Clifford operator and then uses that state for gate teleportation. In the version relevant to this work, the target state is the logical $|H_{XY}\rangle$ state, and the protocol is constructed so that the preparation overhead is reduced relative to previous cultivation proposals. The main benefit of the protocol is that it combines a small initial code, repeated logical checks, and mostly unitary code growth in a way that lowers the expected number of attempts and the total spacetime overhead [1].

The full procedure is divided into three stages referred to as injection, cultivation, and escape. In the injection stage, a logical magic state is prepared on a small rotated surface code. This step is necessarily noisy, since it starts from a physical non-Clifford resource and embeds it into a small code block. In the cultivation stage, that encoded state is transformed into an unrotated surface code, where the fold-transversal logical operator can be measured. For the fault-distance-three protocol, this logical operator is measured twice using a three-qubit GHZ ancilla. After these logical checks, a round of stabilizer measurements is performed and the outcome is decoded. The accepted shots are then passed to the escape stage, where the code is grown to a larger distance using a combination of unitary growth and stabilizer-measurement-based growth, together with further post-selection. In this way, the protocol attempts to gain the error sup-

pression of repeated logical checks without paying the full cost of performing all growth through measurement [1].

For the purposes of neutral atom implementation, the cultivation stage is the most important part. The logical check is not just an abstract measurement; it decomposes into a concrete circuit involving CZ and CCZ gates between the data block and a GHZ ancilla. This is where the architecture enters directly. The FTMSC paper notes that the transformation from the rotated to the unrotated code and the subsequent logical check are most naturally realized on hardware with reconfigurable qubits. Neutral atom arrays are a particularly natural example of such a platform. They can move qubits between different zones, support non-local connectivity, and already admit logical processor architectures with distinct storage, entangling, and readout regions. That means the protocol can be viewed not only as a logical routine but also as a hardware mapping problem: how should one place, move, hide, and entangle atoms in order to realize the logical check efficiently?

This hardware mapping problem has several parts. First, the CCZ gates required by the logical check may have to be implemented on asymmetric atomic layouts, so one needs a method to construct high-quality pulses beyond the symmetric cases usually treated in idealized studies. Second, the protocol contains repeated intermediate measurements, so one needs an efficient method of readout that does not unduly disturb the rest of the array. In a zoned neutral atom architecture, this can be handled by physically separating the readout region from the storage and entangling regions and by moving selected atoms to be measured. More generally, one may also consider hiding or shelving spectator atoms during readout so that the measurement overhead acts only where needed. Third, the total cost of the protocol depends on whether it is better to realize the logical check through native asymmetric CCZ gates or through an entangling-zone architecture that relies more heavily on atom movement and feedforward. These are the implementation questions that motivate the remainder of this work.

In this sense, FTMSC is a useful protocol for studying neutral atom hardware because it forces one to confront several hardware questions at once. It is not only a question of gate fidelity. One must also account for qubit geometry, the time required to complete repeated logical checks, the effect of intermediate readout, and the trade-off between more complex native gates and more complex movement schedules. The protocol therefore provides a concrete setting in which different neutral atom architectural choices can be compared at the logical level.

II. QUBIT LAYOUT

The layout of the data qubits is well established as a diagonal grid with check qubits at the center of each cell. What is not yet established is the position of the

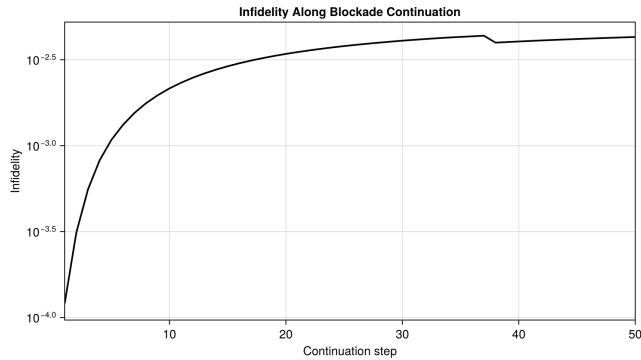


FIG. 2. Infidelity over the course of adiabatic continuation steps with a starting point of $[10, 10, 10]$ $V/\Omega_{\max}$ couplings. Even as the average blockade strength increases, the infidelity continually decreases with growing asymmetry.

GHZ ancilla qubits. When performing the simulation of the logical check phase, we must make a choice about the position of the ancilla in order to choose our CCZ gates appropriately. To make this choice there are several parameters we must keep in mind: degree of asymmetry, blockade strength, and physical constraints of the acousto-optic deflectors (AODs) which keep the atoms in place.

We begin with the degree of asymmetry. It is well known that higher blockade strengths allow for higher-quality pulses; however, the asymmetry of the atom layout is another factor that must be considered. As per Fig. 2, we can see that attempting to remain as symmetric as possible results in higher fidelities. The data qubit layout does not lend itself to perfectly equilateral positions if we are to do an in-place gate, so when picking the ancilla position we try to keep the atoms as close to equilateral as is reasonably achievable.

The next problem to solve is keeping the blockade strength reasonably large. As mentioned before, a strong blockade is required for high-fidelity gates. The relation between distance and blockade strength is taken from Table AIV in [10]. We use the $83s_{1/2}$ orbital here and accept the loss of parallelization that comes with it in order to ensure that our blockade strength is large enough for high-quality gates. Gate times are on the order of $0.3 \mu s$, so the additional $0.6 \mu s$ gained during the cycle time is negligible compared to the overall cycle time of FTMSC and is still faster than an atom-movement architecture (see Section IV).

The final aspect to keep in mind is the physical capabilities of AODs to trap the neutral atoms. AODs are very effective at maintaining a rectangular lattice of atoms in place, but they fail when the atom traps are in too close of proximity to each other; generally the spacing is around $4.5 \mu m$ [11]. One could imagine running an optimization problem with these constraints in mind and trying to find the absolute best positioning for the ancilla. However, we believe this to be too costly for the benefit

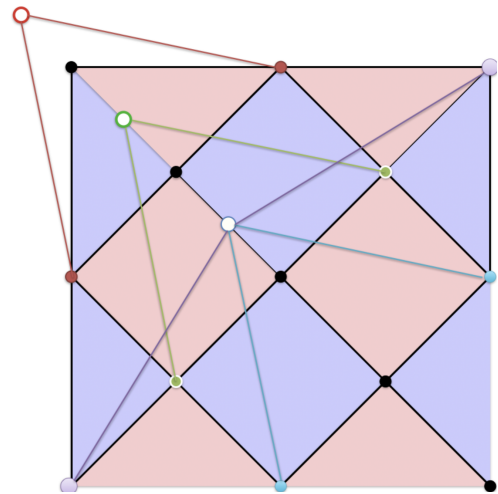


FIG. 3. Layout of ancilla qubits (empty circles) on the logical check circuit. Red, blue, and green indicate short-range gates while purple represents the long-range gate. The lattice spacing is $8 \mu m$.

it would provide and instead settle for an educated guess for the purposes of this work. Fig. 3 shows the final layout chosen for this work. The lattice spacing is doubled in the surface code and the ancilla are placed along the main diagonal to maximize both symmetry and blockade strength while staying consistent with the physical realities of AODs. The blockade strength for the short-range gates is $[40, 20, 20]$ and for the long-range gates is $[7.6, 4.11, 4.11]$, both in units of $B/\Omega_{\max}$.

III. MSC SIMULATION

The main simulation difficulty in this work is that the cultivation check on the logical $|H_{XY}\rangle$ state is not fully Clifford once realistic neutral atom CCZ gates are used. At the same time, most of the rest of the FTMSC protocol is still naturally described by a stabilizer circuit. This means that a full end-to-end density matrix simulation would be unnecessarily expensive, while a pure `stim` simulation would miss exactly the leakage and non-unitarity that we care about most. We therefore use a hybrid simulation strategy: the non-Clifford part of the protocol is simulated exactly with a quantum trajectories method, and the accepted outputs of that exact simulation are then handed off to a `stim` simulation of the later Clifford portion of the protocol. This approach starts from the handoff procedure used in the original FTMSC codebase, but extends it to the case where the CCZ gates are not ideal unitaries and may contain leakage-like behavior [1, 7, 12, 13].

The basic idea is as follows. We first compile the logical preparation and cultivation circuit up to the point where the logical check has been completed. In the ideal FTMSC protocol, this check is expressed using Clifford

operations together with CY-type interactions. In our neutral atom implementation, those check layers are replaced by explicit three-qubit gates built from the asymmetric CCZ pulse solutions found from the Hamiltonian simulation. The short-range and long-range CCZ locations are allowed to use different 8×8 matrices, reflecting the fact that the physical three-atom geometries are not the same. These matrices are generally not exactly unitary, because amplitude can leak out of the computational subspace during the pulse. Instead of forcing such a gate back into a unitary model, we treat the simulated matrix K as a physical Kraus operator. We then complete it into a trace-preserving channel by defining

$$K_0 = K, \quad K_1 = \sqrt{I - K^\dagger K}. \quad (5)$$

This is the minimal completion consistent with the measured or simulated non-unitary action of the gate on the computational subspace.

Once the gate is written in Kraus form, we simulate the cultivation circuit using a Monte Carlo wave-function or quantum trajectory unraveling [12, 13]. For each shot, the simulator samples one Kraus branch for each non-unitary CCZ application and propagates a pure state conditioned on that stochastic history. This is important for two reasons. First, it keeps the exact simulation tractable even when the gate is non-unitary. Second, it preserves the distinction between heralded and unheralded events. A trajectory that leaks population in a way that later triggers post-selection should simply be discarded, whereas a trajectory that survives all checks but leaves the logical qubit slightly rotated should contribute to the logical error rate rather than the discard rate. That distinction would be much less transparent if the entire prefix were compressed into a single effective depolarizing channel.

For each exact trajectory shot, we record whether the post-selection checks from injection and cultivation were passed. If they were not, the shot is discarded at that stage. If they were passed, we extract the reduced state of the logical output qubit and convert it to a logical Bloch vector. We then correct the logical Pauli frame using the measured syndrome parities so that two trajectories which differ only by a tracked Pauli frame are not treated as physically different logical outputs. The accepted shots are binned by the corresponding error-location key and by the resulting logical Bloch vector. This step is the central handoff object in the simulation. It converts the non-Clifford neutral atom dynamics into a classical distribution over accepted logical states attached to stabilizer-compatible error locations.

The later portion of the protocol is then simulated separately using `stim` [14]. At this point the non-unitary CCZ gates no longer appear explicitly. Instead, each accepted error-location class from the exact simulation is mapped onto the corresponding stabilizer circuit for the remainder of the FTMSC protocol, and that stabilizer circuit is sampled many times. We use `stim` for the detector-sampler back end and `PyMatching` v2 for decod-

ing through sparse blossom minimum-weight matching [14, 15]. Operationally, this means that the exact simulation is only paying for the part of the circuit that actually needs non-stabilizer physics, while the much larger Monte Carlo sample used to estimate the logical error rate is pushed into the fast stabilizer regime. This is what makes the larger aggregates in this work feasible.

There is one further detail that is important for interpreting the plotted curves. In the `stim` stage, we do not only record whether a shot is accepted or rejected. We also decode the shot and compute a confidence gap from the difference between the best and second-best matching weights. This gives a confidence-like metric for the decoder output, which we bin in decibel-like units. The final logical error curves are then built by conditioning on a minimum gap threshold. In this way, the plots compare not only the raw logical error rate, but also how the expected number of attempts changes as one asks for more strongly post-selected outcomes. This is the quantity that is most relevant for comparing different CCZ implementations inside FTMSC, since the protocol is intrinsically post-selected.

To compare architectures, we treat atom movement as a separate error channel from the native asymmetric CCZ pulse errors. In other words, the “gate-only” simulations and the “movement” simulations are not fitting the same effect twice. The gate-only runs use the short-range and long-range CCZ pulse models directly on a fixed layout. The movement runs instead add a simple zoned-architecture model in which the qubits involved in the logical check are moved into an entangling zone, kept there for the two logical-check rounds, and then moved back out. This is meant to approximate the storage/entangling/readout zone picture that has now become standard in neutral atom logical processors [4, 16, 17]. The model is deliberately simple. We do not attempt a full routing optimization. Instead, we place dephasing only on the moved qubits during the move-in and move-out windows, and we treat movement-induced loss as a heralded discard that increases the expected number of attempts.

The movement times and error rates are anchored to recent transport benchmarks. In particular, we use the demonstrated coherent transport and trap-transfer results from large-scale tweezer arrays as the physical reference for the transport channel [16]. We then compare the resulting logical-check cycle time to the repeated-round timing reported for a recent fault-tolerant neutral atom architecture [17]. This gives a first-pass architecture comparison that is simple enough to interpret. It is not meant to be the final word on atom shuttling, since a more complete treatment would also include congestion, settling, recooling, and possible gate degradation after movement. However, it is already sufficient to separate two questions that would otherwise be mixed together: how much of the logical error is coming from the native asymmetric CCZ pulse itself, and how much is coming from the extra movement overhead required by a zoned implementation.

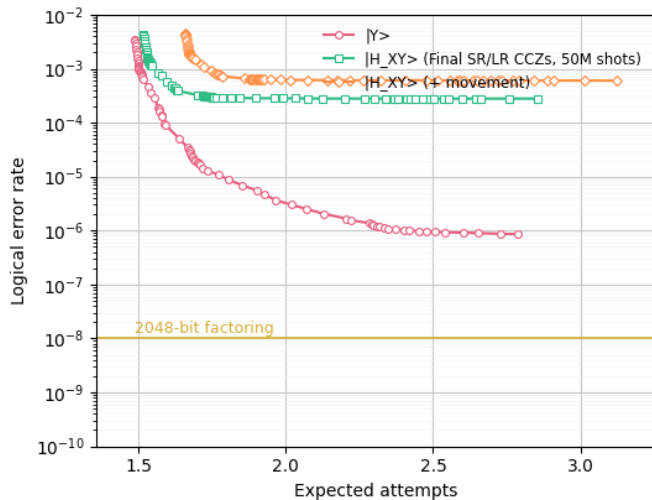


FIG. 4. Full-scale handoff+stim simulation of an in-place (SR/LR) and movement-based architecture. The $|Y\rangle$ curve represents an idealized logical check from [1]. Movement includes an atom loss rate and a dephasing rate. The short-range/long-range regime represents in-place addressing of atoms to implement a CCZ gate.

IV. DATA AND COMPARISONS

Fig. 4 represents the final plot for this work. It shows three curves: an ideal version of the logical check without gate error to serve as a benchmark, an in-place curve using our found short-range (SR) and long-range (LR) gates, and finally a movement-based approach which takes its dephasing rate from [3]. We can see a clear degradation in performance from the ideal curve once any real error is included in our model, which is to be expected. It is clear that the in-place architecture wins in terms of logical error rate (LER), but another important question is the cycle time between the two architectures.

The total process time is determined not only by the entangling pulse time, but also by the time required to move atoms into an entangling zone and to complete the mid-circuit readout and feedforward steps required by the protocol. For the logical check considered in Fold-Transversal MSC, the fault-distance-three implementation requires 21 CZ gates and 4 CCZ gates per check, while the fault-distance-five implementation requires 73 CZ gates and 16 CCZ gates per check. Using a representative neutral atom operating point of $\Omega/(2\pi) = 3.9$ MHz, these correspond to approximate entangling times of $t_{CZ} \approx 0.31 \mu s$ and $t_{CCZ} \approx 0.45 \mu s$. If one further assumes an effective movement layer time of roughly $50 \mu s$ together with a readout and feedforward latency of roughly 1 ms, then a simple scheduling estimate gives a total logical check time of about 1.45 ms for fault distance three and about 1.91 ms for fault distance five.

This timing estimate is important because it shows that, in a zoned architecture, the total cycle time is not limited by the raw entangling pulse durations. The CZ

and CCZ pulses themselves remain sub-microsecond, but the full logical check is pushed to the millisecond scale by movement overhead and especially by the classical loop associated with readout and feedforward. Since the fault-distance-three cultivation stage performs the logical check twice, the logical check portion alone is already on the order of 2.9 ms before including the injection, stabilizer measurement, and escape steps of the full protocol. This suggests that the relevant comparison between our adiabatically found CCZ family and a zoned architecture is not simply one of gate fidelity, but one of full-stack process time.

The comparison is therefore between two different ways of paying the cost of implementing the same logical operation. In the zoned architecture, one pays that cost through atom transport and measurement overhead while using well-controlled entangling regions. In our approach, one attempts to pay that cost through more complicated pulse engineering on the physical gate itself in order to directly realize the asymmetric CCZ gates needed by the protocol. If the resulting non-unitary CCZ gates can be kept at sufficiently low error, then part of the movement overhead of the zoned architecture may be avoided. On the other hand, if the asymmetric pulse family introduces too much leakage or coherent error, then that benefit in cycle time may be offset by worse logical performance inside the MSC routine.

V. FURTHER WORK

The main remaining question is how each architecture, in-place and movement-based, will be affected by increasing the code distance up to $d = 5$. Throughout this work we use a code distance of $d = 3$ for more timely simulations. This assumption, however, is not realistic for actual applications of FTMSC. The goal is to reduce the LER to the 10^{-8} benchmark, which is unachievable for $d = 3$ codes as can be seen from the idealized curve. To reach this point, we will have to run much more expensive simulations which may require additional approximations to remain tractable.

Additionally, realistic noise channels are not yet incorporated directly into the gates. Rydberg decay and atom movement both have the potential to decrease the LER or increase the expected number of attempts once factored into the gates. Again, this will worsen the $d = 3$ plot, but the more interesting question is whether they make the $d = 5$ case similarly error-filled.

Including atom hiding is a promising way to improve this work. Hiding can be used to increase parallelization, reduce overhead from using the tightly focused beams of the in-place architecture, and reduce overhead from ancilla readout.

A final way to improve the simulation is to create a protocol to minimize leakage error, since that is the main error channel causing the in-place architecture to suffer in LER. The long-range gate in particular has non-unitarity

on the order of 10^{-3} . If this leakage can be minimized, that is an immediate win in terms of LER for this procedure.

VI. CONCLUSION

In this work we have set out to make FTMSC work on a realistic neutral atom device. In doing so, we addressed

the problem of asymmetric CCZ gates through the use of an adiabatic pulse engineering method, incorporated simulation of real gates into the handoff using quantum trajectories after making the gates into a Kraus channel, and compared this approach to other possible architectures, namely an atom-shuttling approach, by approximating the loss rate and dephasing rate. We find the in-place architecture using the long-range asymmetric gates to be superior in both LER and total cycle time within the assumptions of the models used in this work.

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